

- 46.** Tank irrigation is commonly found in South-Central parts of India. What could be the reason?

1. Insufficient shallow ground water.
2. Rocky plateau with impervious surface depression.
3. Undulating terrain helps in accumulation of rain water in depression or man-made tank.

Select the correct answer using the codes given below.

- (a) Only 1 (b) 1 and 2
(c) 2 and 3 (d) All of these

- 47.** Which of the following is/are the reason(s) behind Gujarat being the leading producer of Salt in India?

1. The long length of coastline.
2. Long duration of hot and dry conditions.
3. Presence of gulf areas.

Select the correct answer using the codes given below.

- (a) 1 and 2 (b) Only 2
(c) 1 and 3 (d) All of these

- 48.** If 82° 30' East longitude (Allahabad) shows 6:00 am of Sunday (local time), what would be the local time of Florida (USA) located at 82° West longitude?

- (a) 6 : 58 pm of Saturday
(b) 7 : 02 pm of Sunday
(c) 6 : 58 am of Sunday
(d) 7 : 02 am of Saturday

- 49.** Consider the following statement(s) with regard to cold waves in winter season in Northern India

1. There is lack of maritime influence.
2. Northern India is nearer to the Himalayan region.
3. Air mass comes from polar regions of Northern India.

Which of the statement(s) given above is/are correct?

- (a) Only 1 (b) 2 and 3
(c) 1 and 3 (d) 1 and 2

- 50.** Match the following

List I (Sanctuaries)	List II (States)
A. Sharavati Valley	1. Gujarat
B. Satkosia Gorge	2. Goa
C. Pirotan Island	3. Odisha
D. Bhagwan Mahavir	4. Karnataka

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 2 1 3 4 | (b) 4 1 3 2 |
| (c) 4 3 1 2 | (d) 2 3 1 4 |

- 51. Statement I** India's offshore and deep sea fish catch is very poor considering the marine potential.

Statement II Indian coast does not have many gulfs, bays, estuaries and backwaters.

Codes

- (a) Both the statements are individually true and Statement II is the correct explanation of Statement I
(b) Both the statements are individually true, but Statement II is not the correct explanation of Statement I
(c) Statement I is true, but Statement II is false
(d) Statement I is false, but Statement II is true

2015 (II)

- 52.** Match the following

List I (Regions)	List II (States)
A. Baghelkhand	1. West Bengal
B. Kuttanad	2. Madhya Pradesh
C. Saurashtra	3. Kerala
D. Rarh	4. Gujarat

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 2 3 4 1 | (b) 2 4 3 1 |
| (c) 1 3 4 2 | (d) 1 4 3 2 |

- 53.** Match the following

List I (Lakes)	List II (Types)
A. Ashtamudi Kayal	1. Residual (Sweet water)
B. Himayat Sagar	2. Lagoon
C. Dal Lake	3. Glacial
D. Tsomgo Lake	4. Artificial (Sweet water)

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 2 4 1 3 | (b) 2 1 4 3 |
| (c) 3 1 4 2 | (d) 3 4 1 2 |

- 54.** Match the following

List I (Cities)	List II (Earthquake Zones)
A. Kolkata	1. Zone V
B. Guwahati	2. Zone IV
C. Delhi	3. Zone III
D. Chennai	4. Zone II

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 3 2 1 4 | (b) 3 1 2 4 |
| (c) 4 1 2 3 | (d) 4 2 1 3 |

- 55.** Match the following

List I (Farming Areas)	List II (States)
A. Doab	1. Assam
B. Char	2. Karnataka
C. Maidan	3. Punjab
D. Terai	4. Uttar Pradesh

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 3 2 1 4 | (b) 3 1 2 4 |
| (c) 4 1 2 3 | (d) 4 2 1 3 |

- 56.** Match the following

List I (Steel Mills)	List I (States)
A. Kalinganagra	1. West Bengal
B. Vijayanagara	2. Tamil Nadu
C. Salem	3. Odisha
D. Durgapur	4. Karnataka

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 1 4 2 3 | (b) 1 2 4 3 |
| (c) 3 4 2 1 | (d) 3 2 4 1 |

- 57.** Which of the following are the major factors responsible for the monsoon type of climate in India?

1. Location
2. Thermal contrast
3. Upper air circulation
4. The Himalayan Mountains

Select the correct answer using the codes given below.

- (a) 1 and 4 (b) 1, 2 and 3
(c) 2, 3 and 4 (d) All of these

- 58.** Consider the following Indian states

1. Andhra Pradesh
2. Chhattisgarh
3. Tamil Nadu
4. Telangana
5. Uttarakhand

Which among the states given above are the largest and smallest respectively (in terms of geographical area)?

- (a) 3 and 4 (b) 3 and 2
(c) 1 and 5 (d) 2 and 5

- 59.** Match the following

List I (Plants)	List II (Natural Vegetation Types)
A. Ebony	1. Moist deciduous
B. Shisham	2. Himalayan moist
C. Walnut	3. Alpine
D. Birch	4. Tropical evergreen

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 4 1 2 3 | (b) 4 2 1 3 |
| (c) 3 2 1 4 | (d) 3 1 2 4 |

2016 (I)

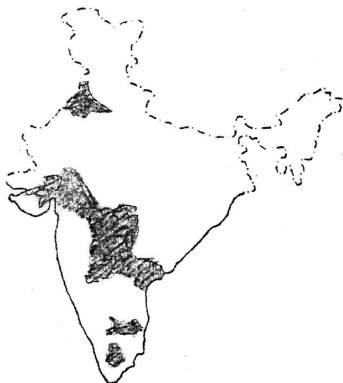
60. Consider the following statement(s) regarding laterite soils of India

1. Laterite soils are generally red in colour.
2. Laterite soils are rich in nitrogen and potash.
3. Laterite soils are well developed in Rajasthan and Uttar Pradesh.
4. Tapioca and cashewnuts grow well in this soil.

Which of the statement(s) given above is/are correct?

- (a) Only 1
- (b) 2, 3 and 4
- (c) 1 and 4
- (d) 1, 2 and 4

61. Consider the following map of India.



The areas marked in the map given above account for the production of which one of the following cash crops?

- (a) Cotton
- (b) Groundnut
- (c) Sugarcane
- (d) Tobacco

62. Pir Panjal range in the Himalayas is a part of

- (a) Shivalik
- (b) Trans Himalaya
- (c) Central Himalaya
- (d) Lesser Himalaya

63. With which one of the following countries, India has signed an MoU under the International Cooperation on Brahmaputra and Sutlej rivers?

- (a) Pakistan
- (b) China
- (c) Bangladesh
- (d) Nepal

64. Headquarters of which one among the following Railway Zones in India is situated at the highest elevation from the mean sea level?

- (a) East–Central Railway
- (b) South–Eastern Railway
- (c) South–Western Railway
- (d) West–Central Railway

65. Consider the following statement(s)

1. Most of the coal and the ferrous group of minerals in India occur in the peninsula, South of the Vindhyas.
2. The peninsular India once formed part of the super-continent which included Australia, Antarctica, Africa and South America.

Which of the statement(s) given above is/are correct?

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

66. The Mahatma Gandhi National Marine Park is located in

- (a) Pirotan Island
- (b) Rameswaram
- (c) Ganga Sagar Island
- (d) Port Blair

67. Which one of the following islands is of volcanic origin?

- (a) Reunion island
- (b) Andaman and Nicobar islands
- (c) Lakshadweep islands
- (d) Maldives

68. Consider the following tributaries of river Brahmaputra

1. Lohit
2. Tista
3. Subansiri
4. Sankosh

Arrange the above rivers from West to East.

- (a) 2, 4, 3, 1
- (b) 2, 3, 4, 1
- (c) 4, 2, 3, 1
- (d) 3, 1, 2, 4

69. Which of the following statement(s) regarding chemical industry in India is/are correct?

1. Chemical industry is one of the oldest industries in India.
2. Dyestuff sector is one of the important segments of chemical industry.
3. Textile industry accounts for the largest consumption of dyestuffs.

Select the correct answer using the codes given below.

- (a) Only 1
- (b) 2 and 3
- (c) 1 and 3
- (d) All of these

70. Arrange the following popular hill stations of India in terms of their height (from highest to the lowest) from the mean sea level.

1. Mussoorie
2. Shimla
3. Ooty
4. Darjeeling

Codes

- (a) 3, 2, 4, 1
- (b) 3, 1, 4, 2
- (c) 2, 3, 4, 1
- (d) 2, 4, 1, 3

71. Which of the following statements relating to the Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 are correct?

1. The Act recognises forest rights of forest dwelling Scheduled Tribes who have been occupying the forest land before 25th October, 1980.
2. The onus of implementation of the Act lies at the level of the State/UT Governments.
3. The Act seeks to recognise and vest certain forest rights in the forest dwelling Scheduled Tribes and other traditional forest dwellers.

Select the correct answer using the codes given below.

- (a) 1 and 2
- (b) 2 and 3
- (c) 1 and 3
- (d) All of these

Practice Exercise

[illegible][illegible]

PART III

ENVIRONMENT, ECOLOGY AND BIODIVERSITY

ENVIRONMENT

- Environment literally means the surrounding and everything that affects an organism during its lifetime, collectively known as its environment.
- In other words, “environment is the sum of the components of water, air and land; inter-relationships among them and also with human beings, other living organisms and property”. It includes all the physical and biological surroundings and their interactions.

The environment broadly consists of two types of components

- Abiotic Components** Abiotic components of environment includes non-living elements like-land, water and air.
- Biotic Components** Biotic components of environment include all the life forms like plants, animals, decomposers.

Types of Environment

There are three types of environment :

- Natural Environment** It consists of all abiotic (non-living) and biotic (living) components, which exist naturally on Earth.
- Man-Made Environment** It consists of those places which are artificially made by man by planned manipulation. It includes crop fields, urban centres, industrial set-ups etc.
- Social Environment** It includes cultural norms and values prevailing in a particular society. It also includes the network of political, economic and religious institutions which often decide the environmental resources to be utilised by the people.

ECOLOGY

It is a science, in which study of organism is undertaken in relation to their environment. This science developed in response to the increasing awareness of inter-relationships between plants, animals and their physical habitats.

Ecosystem

- Then term ecosystem was first used by AG Tansley in 1935, who defined ecosystem as a particular category of physical system, consisting of organisms and inorganic components in a relatively stable equilibrium, open and of various sizes and kinds.

- In an ecosystem, the biotic and abiotic components are inseparably inter-related and interact with each other. Being an open system, an ecosystem has energy and material flowing across its boundaries.

Components of Ecosystem

Components of ecosystem can be classified into two parts, which are as follow :

- Abiotic Part** It is the non-living component, e.g. air, water, soil, suspended particulate matter etc. Abiotic components of an ecosystem includes inorganic substances (carbon dioxide, nitrogen, oxygen water, etc), organic chemicals (chlorophyll etc) and climate factors (light, temperature, humidity etc).
- Biotic Part** It includes plants, animals and micro-organisms. It can be classified into various categories on the basis of functions.

Producers

- Plants are a major part of producers. Apart from trees and plants, producers also include micro-organisms such as bacteria near ocean vents. They produce their food themselves.
- *Producers make their food or derive their energy by two processes :*
 - Photosynthesis
 - Chemosynthesis

Consumers (Heterotrophs)

Organisms that gather energy by consuming organic material from other organisms. Rather than performing photosynthesis or chemosynthesis, consumers perform aerobic respiration, which converts sugar (glucose) and oxygen into carbon dioxide and water to give them energy.

There are four types of consumers, which are as follow :

- Primary Consumers** Organisms that consume mainly producers. Primary consumers are also known as herbivores because they consume mainly plants.
- Secondary Consumers** Organisms that consume mainly primary consumers. Secondary consumers are also known as carnivores because they consume mainly animals.
- Tertiary Consumers** Organisms that consume secondary consumers. Tertiary consumers are to top carnivores because they are not eaten by other animals.
- Decomposer or Detritus** Bacteria, fungi and insects which degrade and decompose organic matter of all types and restore nutrients to the environment.

Functioning of Ecosystem

The main functions of an ecosystem are as follow :

Food Chain

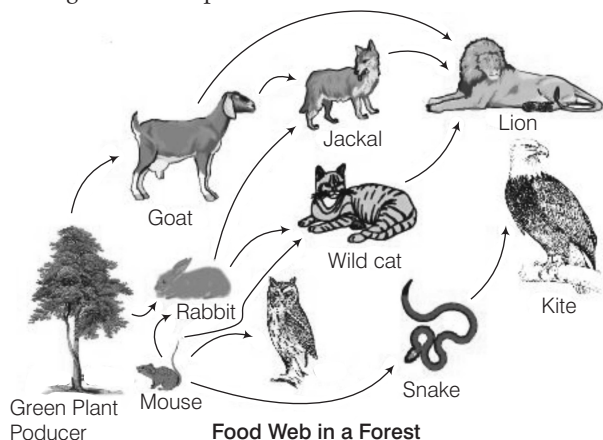
- The feeding of one organism upon another in a sequence of food transfer is known as a food chain. Another definition is that it is the chain of transfer of energy (which typically comes from the Sun) from one organism to another. A simple food chain is like the following

Grass>Insect>Frogs>Snake>Hawk

- Except in deep-sea hydrothermal ecosystems, all food chains start with photosynthesis and end with decay.

Food Web

- A network of food chains or feeding relationships by which energy and nutrients are passed on from one species of living organism to another is called Food Web.
- A food web is represented by a graphical description of feeding relationships among species in an ecological community, e.g. of who eats whom. It is also a means of showing how energy and materials (e.g. carbon) flow through a community of species as a result of these feeding relationships.



- Trophic Levels** Trophic levels are the feeding position in a food chain such as primary producers, herbivore, primary carnivore etc. Generally, green plants form the first trophic level, the producers. Herbivores form the second trophic level, while carnivores and omnivores form the third and even the fourth trophic levels.
- Ecological Pyramids** An ecological pyramid is a graphical representation designed to show the number of organisms, energy relationships and biomass of an ecosystem. They are also called Eltonian pyramids after Charles Elton, who developed the concept of ecological pyramids.
- Pyramid of Biomass** A pyramid of biomass is a representation of the amount of energy contained in biomass, at different trophic levels for a particular time. It is measured in grams per square metre or calories per sq m.

There are two types of biomass pyramids, which are as follow :

- Upright Pyramid** An upright pyramid is one where the combined weight of producers is larger than the combined weight of consumers. e.g. forest ecosystem.
- Inverted Pyramid** An inverted pyramid is one where the combined weight of producers is smaller than the combined weight of consumers. e.g. an aquatic ecosystem.

Pyramid of Numbers

The pyramid of numbers represents the number of organisms in each trophic level.

- Upright, partly upright and inverted are the three types of pyramids of numbers. An aquatic ecosystem is an example of upright pyramid where the number of organisms becomes fewer and fewer higher up in the pyramid.
- A forest ecosystem is an example of a partially upright pyramid, as fewer producers support more primary consumers, but there are less secondary and tertiary consumers.
- An inverted pyramid of numbers is one where the number of organisms depending on the lower levels grows closer toward the apex. e.g. a parasitic food chain.

Pyramid of Energy

The pyramid of energy represents the total amount of energy consumed by each trophic level. An energy pyramid is always upright as the total amount of energy available for utilisation in layer above is less than the energy available in the lower levels. This happens because during energy transfer from lower to higher levels, some energy is always lost. Energy is also lost at every level in the form of heat, respiration etc.

Ecosystem Productivity

The productivity of an ecosystem refers to the rate of production, i.e. the amount of organic matter accumulated per unit area in unit time. It is of following types

- Gross Primary Productivity (GPP)** It is minus energy utilised in respiration.

$$NPP = GPP - R$$
- Net Primary Productivity (NPP)** It is the amount of chemical energy, that is generated by primary producer in a given period of time. It is also defined as rate of production of organic matter. NPP is the available biomass for consumption to heterotrophs.
- Secondary Productivity (SP)** It is the rate of energy storage at consumer level.
- Net Productivity (NP)** It is the rate of storage of organic matter not used by the heterotrophs or consumers.